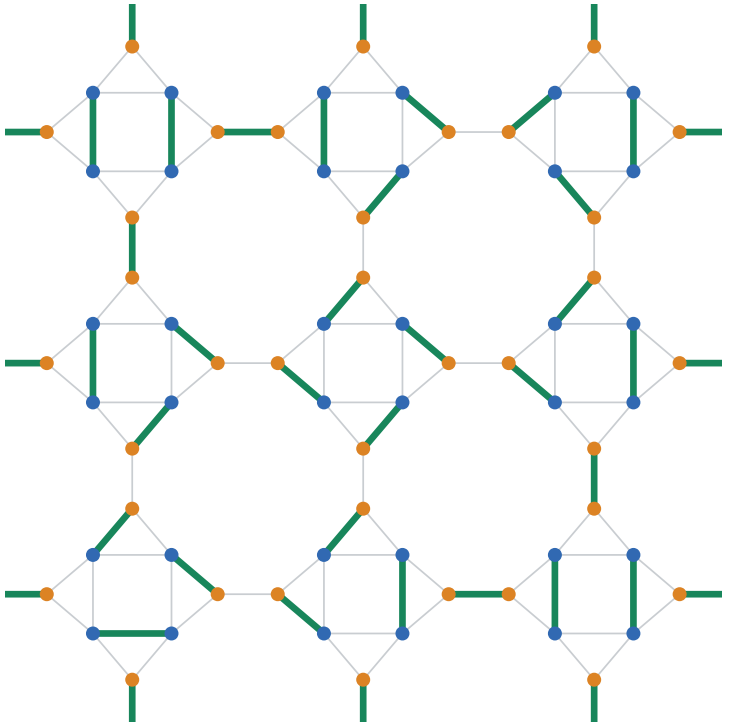
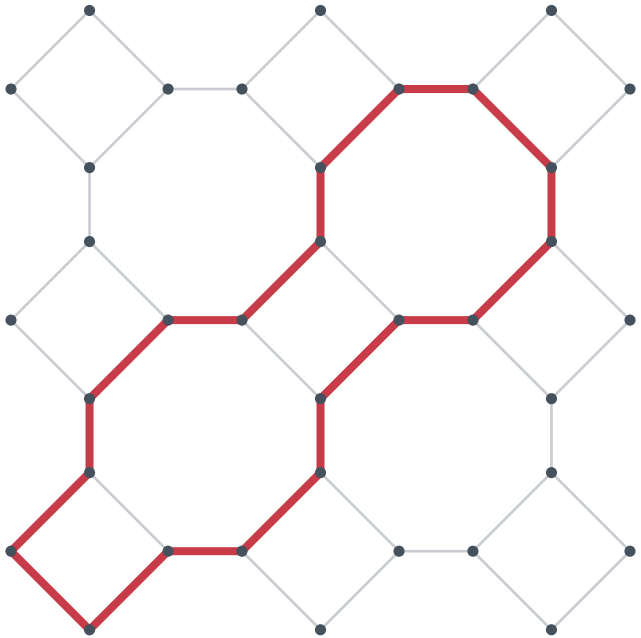
A checked covering-to-loop map; all physical spins remain paired

1. Your decorated spin lattice



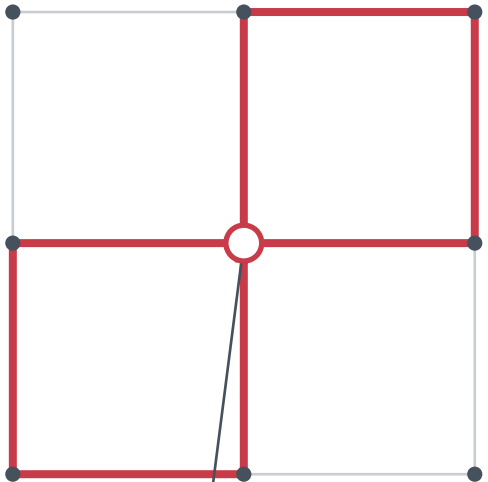
One spin on each square edge
Two spins on each dummy link
One allowed singlet covering

2. Closed loops on square–octagon



Flips relative to one stated reference covering
Every vertex has loop degree 0 or 2

3. Projection onto the square lattice



Four incident loop edges:
retain the local resolution